

International Journal of Number Theory
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The Algebraic Content of Mod-8 Residue Structure in Collatz Stopping Times: A Prefix-Decomposition Conjecture and Bayesian Head-to-Head Replication

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Received (Day Month Year)
 Revised (Day Month Year)
 Accepted (Day Month Year)
 Published (Day Month Year)

We propose a structural origin for the mod-8 residue-class random effect that Bonacorsi and Bordoni (2026) introduce empirically in their hierarchical NB2-GLM for Collatz stopping times. By tracking the symbolic Collatz state through its deterministic prefix, we identify a closed-form per-class structural offset $\alpha_{\text{det}}(r) = \text{prefix_steps}(r) + K_h \cdot \log(a_{\text{final}}(r)/2^k)$, where $K_h = 3/\log(4/3)$ is the textbook Collatz drift coefficient and a_{final} is the prefix's terminal multiplier. We conjecture that the empirical mod-8 random effect equals α_{det} at $k = 3$, rescaled by the GLM's log link with leading-order coefficient $1/\mathbb{E}[\sigma]$. Evidence runs on two sides: a 40-cell empirical verification of α_{det} 's bridge to Tao 2022's (5.15) at slope unity without parameter tuning, and a head-to-head Bayesian replication under Pathfinder VI at $N = 10^7$ plus HMC validation at $N = 10^4$. The HMC arm confirms the conjecture's exhaustion content predictively: the mod-8 random effect's posterior σ_u vanishes once a_{final} at $k = 6$ is in the model.

Keywords: Collatz conjecture; stopping times; prefix decomposition; mod-8 residue structure; hierarchical Bayesian models.

Mathematics Subject Classification 2020: 11B83, 11K31, 62F15, 65C40

1. Introduction

The Collatz map $T(n) = 3n + 1$ if n odd, $n/2$ if n even, generates total stopping times $\sigma(n)$ whose statistical structure has been studied empirically and analytically [4, 6, 7]. For random odd n the standard random-walk heuristic predicts $E[\sigma \mid n] \approx K_h \cdot \log(n)$ where $K_h = 3/\log(4/3) \approx 10.4282$, derived from the assumption that 2-adic valuations along the Syracuse compression are independent and identically distributed $\text{Geometric}(\frac{1}{2})$ [5]. The heuristic captures the leading order; whether σ has an additional residue-class structure beyond the heuristic single-slope model has been the subject of more recent work.

Bonacorsi and Bordoni [1] approach the question using Bayesian methodology.

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Their NB2-GLM with the hierarchical effects on the mod-8 residue-class substantially outperforms mechanistic generators in the out-of-sample predictive log score and the Wasserstein distance at $N \leq 10^7$. Their decomposition

$$\tau(n) = v_2(n) + \sum_j (1 + K(m_j))$$

separates the initial halvings $v_2(n)$, treated as deterministic, from the odd-block dynamics $K(m) = v_2(3m + 1)$, treated as stochastic. The mod-8 random effect is the residue-class adjustment they fit empirically; they identify it as load-bearing for predictive accuracy but do not claim a structural origin.

This paper proposes a structural origin for the mod-8 random effect that emerges symbolically from the tracking of the Collatz iteration. For odd n with residue $r = n \bmod 2^k$, write $n = 2^k \cdot m + r$ and track the Collatz state as $a \cdot m + c$, starting from $(a_0, c_0) = (2^k, r)$. While a remains even, the state’s parity depends only on $c \bmod 2$, and hence on r alone, independent of m . The iteration is therefore uniform in m until a becomes odd, at which point the next state’s parity depends on m and the determinism breaks. We call the deterministic portion the *prefix*. The prefix terminates after at most k odd-step applications (since $a_0 = 2^k$ has 2-adic valuation exactly k , and each odd step is followed by even-step divisions that strictly decrease $v_2(a)$), with terminal value $a_{\text{final}} = 3^j$ for some $j \in \{1, 2, \dots, k\}$. Empirically, the prefix runs in 7–12 steps depending on r at modular resolutions $k \in \{6, 7, 8, 9\}$.

We conjecture that the Bonacorsi-Bordoni mod-8 random effect is, up to a global additive constant, $\alpha_{\text{det}}(r)$ at $k = 3$ rescaled by the GLM’s log link, with leading-order coefficient $1/\mathbb{E}[\sigma]$ on the log-mean scale. Here $\alpha_{\text{det}}(r)$ is the prefix decomposition’s closed-form per-class structural offset on σ -additive scale, defined in §3.2.

The contribution is positioned carefully. The conjecture is supported by evidence on two sides. On the algebraic side (§3–§4), we present the algebraic structure of the prefix decomposition and its empirical predictions for per-class statistics of σ . The headline empirical finding is that the prefix decomposition’s α_{det} predicts per-class mean first-passage time at four arbitrary thresholds $f(N) \in \{N^{2/3}, \sqrt{N} \log N, \sqrt{N}, \sqrt{N}/\log(N)\}$ with regression slope in $[0.9936, 1.0012]$ across 40 verification cells at the textbook constant K_h without recalibration (full artifact paths in Appendix A). On the data side (§5), we present a head-to-head Bayesian replication in the Bonacorsi-Bordoni framework. The replication runs in two arms: Pathfinder variational inference at $N = 10^7$ [8], and HMC at $N = 10^4$ as a structural-validation arm. The HMC arm confirms that B1’s σ_u posterior is real (not a Pathfinder-collapse artifact), that the fixed-effect a_{final} specifications outperform the mod-8 random-effect specification ordinally, and that adding the mod-8 random effect on top of a_{final} at $k = 6$ produces a posterior σ_u that vanishes – the predictive-level confirmation of the conjecture’s exhaustion content.

The Pathfinder VI choice for the $N = 10^7$ arm deserves explicit framing. Production HMC at that scale hit a Stan 2.36 unified-mode multi-chain lockup that prevented full HMC inference; Pathfinder converged cleanly (experiment 06b).

Pathfinder is a quasi-Newton variational approximation that systematically underestimates posterior dispersion in non-Gaussian hierarchical regions, and accordingly collapses σ_u in any specification with a hierarchical random effect [8]. Absolute log-score numbers for hierarchical specifications (B1, B4) under Pathfinder are therefore lower than HMC would produce on the same data; fixed-effect specifications (B0, B2, B3) are not subject to this bias. The HMC-at- $N = 10^4$ arm sidesteps the lookup at the smaller data scale (parallel-chains mode rather than unified mode) and confirms the structural ordering. The structural-empirical relationship this paper conjectures is independent of inference method: the prefix decomposition is an algebraic structure on the Collatz map, and the conjectured correspondence with the mod-8 random effect is a claim about the algebraic content of the fitted random effect, not a claim about predictive performance under a specific inference procedure. HMC at the Bonacorsi-Bordoni production scale is the natural follow-up for tightening absolute log-score precision; the Bonacorsi-Bordoni group is positioned to run that.

The paper is organized as follows. Section 2 places the contribution in the literature, including [7], [6], [1], and the Bayesian inference methodology of [8]. Section 3 defines the prefix decomposition and presents its algebraic structure, including the closed-form prediction $\langle \alpha_{\text{det}} \rangle = \log(6)/\log(4/3)$ and the bridge to Tao (5.15). Section 4 presents the empirical structural findings on per-class statistics of σ at modular resolutions $k \in \{6, 7, 8, 9\}$ and N up to 2^{27} , including the 40-cell verification of the bridge to Tao's (5.15). Section 3.5 states Conjecture 3.1 and Section 5 presents the head-to-head replication under both Pathfinder VI and HMC. Section 6 discusses implications and natural follow-ups. Section 7 concludes.

2. Background and related work

2.1. Collatz stopping times and the random-walk heuristic

The total stopping time $\sigma(n)$ is the number of Collatz steps from n to 1. The standard heuristic $E[\sigma \mid n] \approx K_h \log(n)$ with $K_h = 3/\log(4/3)$ follows from a random-walk argument: the Syracuse compression $n \rightarrow (3n+1)/2^v$ has expected log-multiplier $\log 3 - \langle v \rangle \cdot \log 2 = -\log(4/3)$ when $v \sim \text{Geom}(1/2)$ and the inverse of this drift gives K_h [4, 5].

Empirical departures from the heuristic at finite N have been documented: $\sigma(n)$ records (OEIS A006877), per-class structure visible in σ -vs- $\log(n)$ scatter plots, and finer modular banding under Bayesian hierarchical fits [1]. Whether these departures reflect genuine residue-class structure or finite- N artifacts is the question this paper, like its predecessors, addresses.

2.2. Terras 1976 and the asymptotic prefix structure

Terras [7] established the asymptotic identity for the structure of the residue class k th iteration: $S_k \approx S_0 \cdot 3^{d(k)}/2^k$ where $d(k)$ is the number of odd-step applications

among the first steps k Collatz starting from a residue class mod 2^k . The Terras Lemma 4 gives the decomposition at the level of step counts but does not retain the symbolic state (a, c) past the asymptotic ratio. The prefix decomposition introduced in §3 is a sharpening of Terras Lemma 4: we track the full symbolic state through the deterministic prefix, retain $a_{\text{final}} \in \{3^1, \dots, 3^k\}$ as a closed-form covariate and use it to predict per-class statistics of σ rather than only the asymptotic ratio.

2.3. *Tao 2022 and the leading-term mean trajectory*

Tao [6] establishes that for almost all N , the mean trajectory $T_x(N) = \log(N/x)/\log(4/3) + O(\log^{0.6}(x))$ holds. The leading term $K_h \cdot \log(N/x)$ appears in Tao’s (5.15) and is the asymptotic content of the Syracuse random-walk heuristic at the level of mean trajectories rather than total stopping times. The empirical bridge in §4 connects the prefix decomposition to Tao’s leading term at the per-residue class level: the closed form $s_{\text{mean}}(r; f) \approx \alpha_{\text{det}}(r) + K_h \cdot \log(N/f(N))$ holds across modular resolutions, data scales, and threshold choices, with α_{det} providing the per-class structural offset and $K_h \cdot \log(N/f(N))$ recovering Tao’s leading term.

2.4. *Bonacorsi and Bordoni 2026*

Bonacorsi and Bordoni [1] develop a hierarchical Bayesian NB2-GLM for Collatz stopping times. Their model includes a random effect of the residue-class of mod-8; the choice of mod-8 was empirically motivated by visible banding in σ -vs- $\log(n)$ plots stratified by residue. Their decomposition $\tau(n) = v_2(n) + \sum_j (1 + K(m_j))$ treats $v_2(n)$ as deterministic and the sequence $K(m_j)$ as stochastic, with the mod-8 random effect capturing the fit residue-class adjustment. They report that the NB2-GLM substantially outperforms mechanistic generators on out-of-sample log score and Wasserstein distance at $N \leq 10^7$ under HMC inference.

This paper proposes that the empirical mod-8 banding has a structural origin in the prefix decomposition. The prefix decomposition extends $v_2(n)$, which Bonacorsi-Bordoni treat as the entire deterministic part, by continuing the symbolic computation through subsequent odd $\leftrightarrow$ even compressions. The full deterministic prefix terminates at $a_{\text{final}} \in \{3^1, \dots, 3^k\}$, and we conjecture that their mod-8 random effect is the empirical realization of $\alpha_{\text{det}}(n \bmod 8)$ at $k = 3$, rescaled by the GLM’s log link with leading-order coefficient $1/\mathbb{E}[\sigma]$.

2.5. *Chang 2026 and the orbit-mixing reduction*

Chang [2] develops a finite-depth structural framework for the compressed Syracuse map and reduces the Collatz conjecture to a one-bit orbit-mixing condition at modular resolution 32. The framework decomposes the burst-gap binary indicator process $X_t = \mathbf{1}[v_2(3n_t + 1) \geq 2]$ into block frequencies and bounds the block-TV distance to a Bernoulli product law at depths $K = 3, 4, 5$. The Map Balance Theorem establishes that among the $2^{K-3} - 1$ burst residues modulo 2^K initiating gaps,

the counts mapping to gap starts $\equiv 3 \pmod{8}$ versus $\equiv 7 \pmod{8}$ differ by exactly 1 for every $K \geq 5$, with the dominant $n \equiv 1 \pmod{8}$ class perfectly balanced. The remaining open problem in the framework is a deterministic balance of bit 4 along a sparse subsequence of burst-ending times.

Chang’s decomposition operates on the block frequencies of a binary indicator process derived from the Collatz dynamics; the prefix decomposition introduced in this paper operates on the symbolic algebraic state $a \cdot m + c$ of the Collatz iteration itself. The two frameworks address different objects and different questions: Chang’s targets the resolution of the conjecture through orbit-level mixing; the present work characterizes the per-class statistics of the total stopping time σ on residue classes $\text{mod } 2^k$ and conjectures their algebraic content. Both frameworks identify the mod-8 residue structure as load-bearing, but for distinct statistical questions: Chang for the gap-length distribution in burst-gap traditions [2, Lemma 4.1], and the present work for the random effect of [1] in their hierarchical NB2-GLM.

2.6. Pathfinder variational inference

Zhang et al. [8] introduce Pathfinder, a parallel quasi-Newton variational inference method for Bayesian models. Pathfinder constructs a Gaussian approximation along the L-BFGS optimization path, importance-resamples from the approximation, and aggregates across parallel paths. It is fast, well-validated for fixed-effect Bayesian regression, and converges cleanly on hierarchical models with the limitation that it systematically underestimates posterior dispersion in non-Gaussian hierarchical regions. The σ_u parameter in hierarchical specifications collapses under Pathfinder relative to its HMC posterior, with corresponding effects on absolute predictive log scores in such specifications.

The head-to-head replication in §5 uses Pathfinder VI for its $N = 10^7$ arm because production HMC on the Bonacorsi-Bordoni model and data scale hit a Stan 2.36 unified-mode multi-chain lockup that prevented full HMC inference. The structural ordering is then validated under HMC at $N = 10^4$ in a separate arm using parallel-chains mode (which sidesteps the lockup at smaller scale). Pathfinder is appropriate for the $N = 10^7$ arm’s structural-identification scope; the HMC arm at $N = 10^4$ confirms the ordering and Conjecture 3.1’s exhaustion content under proper inference. HMC at the Bonacorsi-Bordoni production scale is the natural follow-up for absolute log-score precision and is recommended.

3. The prefix decomposition

3.1. Symbolic state and the deterministic prefix

For an odd integer n with residue $r = n \bmod 2^k$, write $n = 2^k \cdot m + r$. Track the symbolic state through the Collatz iteration as $a \cdot m + c$, starting from $(a_0, c_0) = (2^k, r)$. The state transitions are determined by the parity of $a \cdot m + c$; for the prefix region the parity is determined by the parities of a and c jointly:

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- If a is even and c is even: the state of $a \cdot m + c$ is always even, halve $\rightarrow (a/2, c/2)$
- If a is even and c is odd: the state is always odd, apply $3n+1 \rightarrow (3a, 3c+1)$
- If a is odd: parity depends on m , terminate the prefix

Each odd-step application multiplies a by 3; subsequent even-step applications halve a until it again becomes odd or until the orbit reaches the prefix’s natural termination.

The iteration terminates at the first step where a becomes odd. The initial value $a_0 = 2^k$ contains exactly k factors of 2, so the iteration runs for at most k odd-step applications before a becomes odd. The terminal value a_{final} is therefore of the form 3^j , where j is the number of odd-step applications consumed during the prefix and $1 \leq j \leq k$.

The number of distinct a_{final} values is at most k . Empirically, all k values are realized for $k \in \{4, 5, 6, 7, 8, 9, 10, 11, 12\}$, so the 2^{k-1} odd residue classes $\text{mod } 2^k$ collapse onto exactly k distinct prefix-terminal types. We refer to this as the prefix-collapse structure: residue classes that share the same a_{final} share the same prefix-terminal type.

3.2. The α_{det} prediction

The per-class intercept of σ vs. $\log(n)$ is predicted by

$$\alpha_{\text{det}}(r) := \text{prefix_steps}(r) + K_h \cdot \log(a_{\text{final}}(r)/2^k)$$

where $\text{prefix_steps}(r)$ is the number of Collatz steps applied during the prefix iteration until a becomes odd and $K_h = 3/\log(4/3)$. The prediction holds up to a single global additive constant absorbed into the global intercept μ_α . We refer to this as the closed form α_{det} .

3.3. Closed-form derivation of $\langle \alpha_{\text{det}} \rangle$

The class-averaged α_{det} admits a closed form. With the heuristic that the prefix terminates at $a_{\text{final}} = 3^j$ with probability 2^{-j} (from j independent odd-step Bernoulli events), the expectation is $\langle \alpha_{\text{det}} \rangle = \log(6)/\log(4/3)$ derived exactly.^a The closed form was empirically verified against the per-class average α_{det} at $k \in \{6, 7, 8, 9\}$ to within the sampling precision.

3.4. Bridge to Tao 2022 (5.15)

The prefix decomposition’s α_{det} extends to predict the per-class mean first-passage time at arbitrary thresholds $f(N)$, not only the σ -intercept. The closed form is

$$s_{\text{mean}}(r; f) \approx \alpha_{\text{det}}(r) + K_h \cdot \log(N/f(N))$$

^aDerivation file: `close_form_findings.md` in the repository (see Appendix A).

The right-hand side recovers Tao’s (5.15) leading term $K_h \cdot \log(N/f(N))$ plus a per-class structural offset $\alpha_{\text{det}}(r)$ [6]. Section 4.3 presents the empirical verification of this closed form across 40 verification cells.

3.5. The conjecture

Bonacorsi and Bordoni [1] introduce a mod8 residue-class random effect $u_{n \bmod 8}$ in their NB2-GLM. We conjecture:

Conjecture 3.1 (Prefix-decomposition identification of the mod8 random effect). *At modular resolution $k = 3$ of the Collatz prefix, the per-residue effect on the total stopping time σ is $\alpha_{\text{det}}(n \bmod 8) = \text{prefix_steps}(n \bmod 8) + K_h \cdot \log(a_{\text{final}}(n \bmod 8)/2^k)$ up to a global additive constant. The corresponding random effect $u_{n \bmod 8}$ in the Bonacorsi-Bordoni NB2-GLM with log link is therefore $\alpha_{\text{det}}(n \bmod 8)$ rescaled by the GLM’s link function, with leading-order coefficient $1/\mathbb{E}[\sigma]$ on the log-mean scale.*

Conjecture 3.1 is supported by both the algebraic structure of the prefix decomposition (§4) and a head-to-head Bayesian replication under Pathfinder VI plus HMC validation at $N = 10^4$ (§5) – but is not proven. The conjecture’s content is that the empirical random effect of the Bonacorsi-Bordoni fit has a closed-form algebraic origin in the prefix decomposition’s per-class structural offset α_{det} , which is determined symbolically from $r \bmod 2^k$. The two-scale distinction is essential: $\alpha_{\text{det}}(r)$ is the per-class additive offset on the σ -additive scale (the natural scale for §4’s per-class regressions of σ on $\log(n)$), and u_r is an additive shift on the log-mean of σ in the NB2-GLM’s log link (the scale of §5’s head-to-head). The leading-order link translation $u_r \approx \alpha_{\text{det}}(r)/\mathbb{E}[\sigma]$ comes from the first-order Jacobian of the log link evaluated at the class mean. If the conjecture is correct, the random effect can be replaced by a fixed effect on α_{det} , which has a logarithmic rather than linear parameter count in the modular resolution: at modular resolution k there are exactly k distinct a_{final} values (and hence at most $k + 1$ distinct α_{det} values, one per prefix-step count), versus 2^{k-1} odd residue classes.

4. Empirical structural findings on per-class σ statistics

The findings in this section are about σ on odd integers and the prefix decomposition, presented as empirical regularities verified at multiple modular resolutions and data scales. No theorems are claimed beyond the algebraic content of §3.1 (the prefix iteration terminates at $a_{\text{final}} \in \{3^1, \dots, 3^k\}$). The results below establish that the prefix decomposition’s α_{det} predicts per-class mean σ , per-class variance, per-class first-passage times, and the bridge to Tao (5.15) without parameter tuning, across modular resolutions $k \in \{6, 7, 8, 9\}$ and N up to 2^{27} .

4.1. Per-class intercept $\alpha(r)$ at modular resolutions $k \in \{6, 7, 8, 9\}$

At modular resolution $k = 6$ with $N = 2^{25}$, hierarchical Bayesian fit of

$$\sigma \sim \alpha(r) + \beta(r) \cdot \log(n) + \epsilon$$

with class-level α and β yields per-class α posterior means spanning -27 to $+35$ with $\tau_\alpha \approx 11.7$. Linear regression of these against $\alpha_{\text{det}}(r)$ gives $\alpha_{\text{actual}}(r) = -2.66 + 0.986 \cdot \alpha_{\text{det}}(r)$, $R^2 = 0.9996$, with $\text{SD}(\alpha_{\text{stoch}})/\text{mean}(\alpha \text{ posterior SE}) = 0.18$. The per-class residual is below the noise floor of posterior sampling alone – the prefix prediction over-explains the per-class intercept relative to what posterior sampling noise at the chosen Stan configuration would produce. Extension to higher modular resolutions at $N = 2^{27}$ via direct OLS (per-class regression rather than hierarchical Stan, to avoid the posterior-geometry difficulties at large k) is reported in Table 1.

Table 1. Extension to higher modular resolutions at $N = 2^{27}$.

k	mod	classes	per-class n	R^2	SD(resid) / mean(SE)
6	64	32	2.10M	0.9967	0.96
7	128	64	1.05M	0.9942	0.99
8	256	128	524k	0.9918	0.91
9	512	256	262K	0.9851	0.93

The R^2 declines with k as per-class samples shrink but the signal-to-noise ratio $\text{SD}(\text{resid}) / \text{mean}(\text{SE})$ stays in the band 0.91-0.99 across all four resolutions: residuals scale with sampling noise, with no detectable signal above it. A controlled comparison at $k = 8$ and $N = 2^{27}$ (524K per class) versus the original $k = 6$ and $N = 2^{25}$ analysis (524K per class) gives $R^2 = 0.9918$ and 0.9907 respectively. The same data per class produces the same R^2 regardless of modular resolution.

4.2. The prefix determines all moments by a_{final} clustering

The hypothesis initially tested was full distributional universality of $S(n) = \sigma(n) - \alpha(r) - \beta \cdot \log(n)$ – that $S(n)$ would be class-independent. We found that it is not: per-class variance differs by $\sim 30\%$ relative across classes mod 64; per-class skewness and kurtosis also vary. However, each higher moment, when computed per class and plotted against α_{det} (or equivalently against $\log(a_{\text{final}})$, since these are collinear at fixed k), shows the prefix-prediction pattern:

The clustering is sharper than the Pearson correlations alone indicate: when per-class variance is plotted against α_{det} , classes with the same a_{final} produce indistinguishable variance – they collapse onto a single point. At $k = 6$ with 32 classes, there are exactly 6 distinct points corresponding to $a_{\text{final}} \in \{3^1, \dots, 3^6\}$. The same is true for kurtosis. Skewness shows slightly more within-cluster scatter but the same discrete structure.

Table 2. Pearson correlation with the prefix

k	classes	per-class n	$r(\text{predicted-}\alpha, \text{Var})$	$r(\text{predicted-}\alpha, \text{Kurt})$	$r(\text{predicted-}\alpha, \text{Skew})$
6	32	2.10M	0.99989	0.9497	0.8716
7	64	1.05M	0.99977	0.9337	0.8442
8	128	524K	0.99969	0.9236	0.8250
9	256	262K	0.99943	0.9077	0.7914

This is the empirical content of the prefix-collapse: at modular resolution k , the 2^{k-1} odd residue classes' σ distributions parameterize on a_{final} alone. The full conditional distribution $\sigma(n) \mid r$ is parameterized entirely by $a_{\text{final}}(r)$, with the closed-form predictor $\alpha_{\text{det}}(r)$ recovering not only the mean but all higher central moments at the level of cluster identification.

4.3. The bridge to Tao (5.15): 40-cell verification

α_{det} was constructed as a closed-form prediction for the σ -intercept. It also predicts an independent functional trajectory without recalibration: the mean first-passage time per-class below an arbitrary threshold $f(N)$.

For odd n , let $s(n; f)$ denote the number of Collatz steps from n until the orbit first attains a value $\leq f(N)$, and let $s_{\text{mean}}(r; f)$ be the mean of $s(\cdot; f)$ over $n \equiv r \pmod{2^k}$. Five observables were tested: σ (full descent to 1, equivalent to $f(N) = 1$) and first-passage to $f(N) \in \{N^{2/3}, \sqrt{N} \cdot \log N, \sqrt{N}, \sqrt{N}/\log N\}$. Four modular resolutions: $k \in \{8, 10, 12, 14\}$. Two data scales: $N \in \{2^{25}, 2^{27}\}$. Total: 40 verification cells.

For every (observable, k , N) cell, the regression $s_{\text{mean}}(r) \sim a + \text{slope} \cdot \alpha_{\text{det}}(r)$ with α_{det} built at the textbook K_h gives:

Table 3. 40-cell verification of α_{det} 's prediction of per-class mean trajectory dynamics against Tao 2022 (5.15)

measure	range across all 40 cells
Spearman ρ between α_{det} and s_{mean}	≥ 0.99 in every cell, exactly 1.0 at $k = 8$
slope at K_h , raw mean	$[0.9936, 1.0012]$ – within 0.5% of 1 in every cell
slope at K_h , 1%-trimmed mean	$[0.9808, 0.9944]$

$K_h = 3/\log(4/3) \approx 10.4282$ is the textbook Collatz drift coefficient, derived from first principles and used here as a fixed input rather than estimated from the data; it appears in Tao's (5.15) and predates this paper. The slope is unity at the textbook K_h with no fit calibration. The closed-form prediction $s_{\text{mean}}(r; f) \approx \alpha_{\text{det}}(r) + K_h \cdot \log(N/f(N))$ holds across modular resolutions, data scales, and threshold choices. The right-hand side recovers Tao's (5.15) leading term plus a per-class structural offset $\alpha_{\text{det}}(r)$. The result is the per-residue-class

realization of Tao’s almost-all- N statement. The gap between Tao’s leading term $K_h \cdot \log(N/f(N))$ and the per-class decomposition is structural rather than a finite- N or finite- k artifact. For σ , gaps are stable to within 0.06 across $k = 8 \rightarrow 14$ and within 0.05 across $N = 2^{25} \rightarrow 2^{27}$; the other per-class statistics show comparable stability. With light trimming (top 1% per class), the gap at $\sqrt{N}$ drops to sub-percent of the Tao prediction across every cell, since trimming removes the right-tail contribution Tao’s exceptional-set theorem allows for.

4.4. *Slope universality on σ*

For completeness, we note that the per-class slope $\beta(r)$ of σ vs $\log(n)$ is consistent with the universal value $K_h = 10.4282$ across all classes at all tested N . The moment-of-moments estimator $\hat{\tau}_\beta^2 = \max(0, \text{observed}^2 - \text{SE}^2)$ hits its zero floor by construction at every sample size $N \in \{2^{20}, 2^{22}, 2^{23}, 2^{24}, 2^{25}\}$ on data mod64. The defensible claim is that we cannot reject $\tau_\beta = 0$ at any tested N . The per-class slope universality is consistent with Bonacorsi-Bordoni’s choice to model the slope coefficient as fixed-effect rather than hierarchical.

The pooled empirical slope μ_β oscillates around 10.4282 with amplitude ≈ 0.01 over $N \in [2^{25}, 2^{32}]$, with peak deviation near $N \approx 2^{27}$ and not visibly damping by 2^{32} . The oscillation does not affect per-class structure and is not pursued further here.

4.5. *Tail-shape universality*

Per-class GPD tail-shape parameters $\xi(r)$ trend toward zero from below as N grows: $\langle \xi \rangle$ across classes goes from -0.083 at $N = 2^{20}$ to -0.028 at $N = 2^{25}$. The sub-exponential cliff drop visible in residual CCDFs at fixed N is a finite- N truncation artifact – the limiting tail is exponential. This is consistent with Bonacorsi-Bordoni’s use of negative binomial likelihood which has exponential tails in its log-rate parameterization.

4.6. *Finite- N behavior of μ_β and per-octave structure*

The pooled-OLS slope μ_β approaches the heuristic $K_h = 3/\log(4/3) \approx 10.4282$ non-monotonically. Cumulative streaming OLS at $N \in \{2^{20}, 2^{22}, \dots, 2^{32}\}$ shows μ_β rising from 10.3723 at $N = 2^{20}$ through 10.4293 at $N = 2^{27}$, peaking at 10.4298 at $N = 2^{28}$, then drifting back to 10.4187 at $N = 2^{32}$, with oscillation amplitude ≈ 0.01 not visibly damping at the largest tested scale (Table ??). The crossing of K_h occurs near $N \approx 2^{27}$.

Per-octave OLS restricted to odd $n \in [2^j, 2^{j+1}]$ gives β_{local} values consistently above K_h across all octaves $j \in [17, 26]$, ranging from 10.49 to 10.89 with a peak at $j = 21 - 22$. The cumulative μ_β is a leverage-weighted average of these local slopes against the global K_h .

The mechanism for the per-octave shift is partial. Trajectory-mean $E[v]$ under the Syracuse map is consistently 1.989 to 1.995 across octaves – slightly below the

Geometric(1/2) prediction of 2, though within 0.5%. Substituting these values into the random-walk formula $K(u) = (1+u)/(u \log(2) - \log(3))$ gives K_{pred} ranging from 10.55 to 10.67, fully explaining the systematic baseline shift but not the residual per-octave variation in β_{local} (range 0.40). The residual likely reflects higher-moment trajectory effects (the $v = 4$ and $v = 10$ spikes shift the MGF $M(s) = E[2^{sv}]$ near $s = \theta - 1$ in ways $E[v]$ alone does not capture), step-to-step correlations between consecutive Syracuse iterates, or octave-dependent descent geometry; isolating these is deferred.

We note that the unconditional trajectory measure on v_2 for $qx + 1$ at $q \in \{5, 7, 9, 11\}$ matches Geometric(1/2) to 0.5% on mean and variance [3, 2026-05-02 entry]. The per-octave deviation in $E[v]$ is therefore a feature of the trajectory measure on natural-density-sampled odd integers; not a failure of the Geometric(1/2) baseline at marginal level.

The slope-universality claim of this paper, that $\tau_\beta = 0$ to the noise floor across modular resolutions $k \in \{6, 7, 8, 9\}$ at every N tested, is unaffected by the global μ_β oscillation. The $\tau_\beta = 0$ result is *per-class within fixed N* : classes within a single N show no detectable slope variation. The $\mu_\beta(N)$ behavior is an aggregate property of the natural-density distribution that does not partition into per-class structure.

5. Head-to-head with Bonacorsi-Bordoni NB2-GLM

5.1. Head-to-head methodology

We replicate the Bonacorsi-Bordoni NB2-GLM in Stan with the following specifications:

- B0: $\log(n)$ only (3 parameters)
- B1: B0 + random effect on $(n \bmod 8)$, the Bonacorsi-Bordoni setup (12 parameters)
- B2: B0 + fixed effect on a_{final} at $k = 3$ (6 parameters)
- B3: B0 + fixed effect on a_{final} at $k = 6$ (9 parameters)
- B4: B3 + random effect on $(n \bmod 8)$ (18 parameters)

Priors are matched to Bonacorsi-Bordoni: $\beta \sim \text{Normal}(0, 5)$, $\phi \sim \text{LogN}(\log 5, 1)$, $\sigma_u \sim \text{HalfNormal}(0, 2)$. The head-to-head is run at two scales under two inference procedures.

Pathfinder VI at $N = 10^7$. cmdstanpy 1.3.0 Pathfinder [8], 4 paths $\times$ 1000 importance-resampled draws, on a 500K-observation training subsample with a 50K held-out test set. Production HMC at this scale hit a Stan 2.36 unified-mode multi-chain lockup that pinned all threads at full CPU with zero sampling progress (a synchronization issue with `num_chains=N` in shared-thread-pool mode, distinct from the well-known `reduce_sum` grainsize pathology); Pathfinder converged cleanly. Pathfinder is a quasi-Newton variational approximation that systematically underestimates posterior dispersion in non-Gaussian hierarchical regions [8]. The σ_u parameter therefore collapses under Pathfinder in specifications with a hierarchical

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random effect, and absolute held-out log scores for those specifications (B1, B4) are pulled lower than HMC would produce on the same data. Fixed-effect specifications (B0, B2, B3) are not subject to this bias.

HMC at $N = 10^4$. To validate the structural ordering and Conjecture 3.1’s predictive content under proper inference, we re-ran B0–B4 at $N = 10,000$ under HMC (cmdstanpy 1.3.0, 4 chains $\times$ 2000 iterations each, parallel-chains mode – not unified mode – which sidesteps the Stan 2.36 lockup at this data scale). Train/test split: 8000/2000. All five specifications converged with $\hat{R} < 1.01$, ESS-bulk > 600 , zero divergent transitions, zero treedepth saturations, and E-BFMI > 0.80 . Evaluation metric: log score on the held-out test set. The HMC arm’s purpose is structural validation – the predicted ordinal pattern across specifications and the predicted exhaustion behavior of σ_u in B4 – not absolute log-score numbers, which are not directly comparable across the two scales.

HMC validation at the Bonacorsi-Bordoni production scale ($N = 10^7$ on their infrastructure, where the Stan lockup does not apply) is the natural follow-up for tightening absolute log-score precision; the structural-empirical content of this section’s findings is independent of the inference procedure and is supported by both arms.

5.2. Results

Table 4. Bayesian replication in the Bonacorsi-Bordoni framework, two arms.

Spec	Description	Params	Pathfinder, $N = 10^7$		HMC, $N = 10^4$	
			log score	σ_u	log score	σ_u
B0	log(n) only	3	−274,150.3	—	−10,929.74	—
B1	+ RE on $(n \bmod 8) \leftarrow$ B&B’s setup	12	−274,138.4	0.003*	−10,901.63	0.097
B2	+ FE on a_{final} at $k = 3$	6	−273,288.4	—	−10,900.50	—
B3	+ FE on a_{final} at $k = 6$	9	−272,435.3	—	−10,892.17	—
B4	B3 + RE on $(n \bmod 8)$	18	−272,438.4	0.009*	−10,892.27	0.0075
B&B reported (HMC, full data)	NB2-GLM	—	−272,911.95	—	—	—

* Pathfinder-collapsed; HMC posterior is larger. The HMC-at- $N = 10^4$ arm is the validation; absolute log scores are not comparable across the two scales (test sets differ by $25\times$). All HMC fits passed diagnostics ($\hat{R} < 1.01$, ESS-bulk > 600 , 0 divergences, 0 treedepth saturations, E-BFMI > 0.80).

5.3. What the table says, structurally

The HMC arm validates the Pathfinder caveat. Under Pathfinder at $N = 10^7$, σ_u in B1 reports 0.003 and B1’s log score is essentially indistinguishable from B0’s; the literature on Pathfinder [8] predicts this collapse in non-Gaussian hierarchical regions. Under HMC at $N = 10^4$, B1’s σ_u posterior mean is 0.097 – roughly

32 $\times$ the Pathfinder estimate – and B1’s log score moves 28 nats above B0’s. The mod-8 random effect is real under proper inference; the Pathfinder collapse hypothesis is confirmed quantitatively.

Ordinal structure: B3 > B2 > B1 under both arms. Both Pathfinder at $N = 10^7$ and HMC at $N = 10^4$ rank the specifications identically: the closed-form covariate a_{final} at $k = 3$ (B2) outperforms the mod-8 random effect (B1), and finer modular resolution a_{final} at $k = 6$ (B3) outperforms both. Under HMC, B3 leads B1 by 9.5 nats and B2 leads B1 by 1.1 nats on a 2000-observation test set; the B3-vs-B2 gap (8.3 nats) is itself larger than the B2-vs-B1 gap, indicating that finer modular resolution captures residue-class structure that a_{final} at $k = 3$ alone does not.

Exhaustion: B4 vs B3 (HMC). Adding the mod-8 random effect on top of a_{final} at $k = 6$ (B4) gives an HMC posterior σ_u of 0.0075 – a factor of 13 below B1’s 0.097 – and the B4 log score is statistically indistinguishable from B3’s (B3 better by 0.10 nats with 9 additional random-effect parameters that do not move from zero). The mod-8 random effect, demonstrably real in B1, evaporates once a_{final} at $k = 6$ is in the model. This is the predictive-level confirmation of the conjecture’s exhaustion content: the closed-form covariate at sufficient modular resolution captures the structural information the random effect was empirically tracking.

Magnitude: B2 coefficient against the link-translated prediction. Conjecture 3.1 (§3.5) predicts that u_r on the GLM log-mean scale equals $\alpha_{\text{det}}(r)/\mathbb{E}[\sigma]$ to leading order. With $\mathbb{E}[\sigma] = 154.43$ on the HMC-at- $N = 10^4$ training set and $K_h = 10.4282$, the predicted GLM-scale slope on $\log(a_{\text{final}})$ is 0.0734 when the full α_{det} form is used, or 0.0675 when only $\log(a_{\text{final}})$ is used. B2’s posterior coefficients on the three non-reference a_{final} levels imply an empirical OLS slope on $\log(a_{\text{final}})$ of 0.0774 with 95% credible interval [0.0676, 0.0869]. The full α_{det} prediction lands comfortably inside the empirical interval (mean absolute error per dummy 0.019); the $\log(a_{\text{final}})$ -only prediction lands at the lower edge (MAE 0.032, 40% larger).

The two predictions differ by the $\text{prefix_steps}(r)$ term in $\alpha_{\text{det}}(r) = \text{prefix_steps}(r) + K_h \cdot \log(a_{\text{final}}(r)/2^k)$. Across odd residues mod 8 at $k = 3$, $\text{prefix_steps}(r) = j(r) + k$ exactly, so prefix_steps and $\log(a_{\text{final}})$ are perfectly collinear (Pearson $r = 1.0$); the empirical slope on $\log(a_{\text{final}})$ alone necessarily absorbs both contributions. The $\sim 6\%$ residual between the empirical 0.0774 and the $\log(a_{\text{final}})$ -only prediction 0.0675 is exactly the $1/(\mathbb{E}[\sigma] \log 3)$ contribution prefix_steps would supply, which is precisely what the full α_{det} form recovers. The empirical signal traces to the full α_{det} , consistent with the algebraic content claim of §3.5 once the prefix-steps term is included.

Pathfinder margins partly reflect σ_u collapse. The Pathfinder 850-nat margin of B2 over B1 at $N = 10^7$ partly reflects the σ_u collapse in B1 documented

above; the HMC arm's 1.1-nat B2-over-B1 margin (on a $25\times$ smaller test set) is the proper-inference figure at this scale. The structural ordering survives both arms; the magnitude of any specific Pathfinder margin should be interpreted with the collapse caveat in mind.

5.4. *What the head-to-head does and does not show*

The head-to-head under both inference arms shows: the prefix decomposition's α_{det} is empirically a stronger covariate than the mod 8 random effect at the same modular resolution; finer modular resolution ($k = 6$) outperforms coarser ($k = 3$); and the mod-8 random effect's posterior σ_u collapses once a_{final} at $k = 6$ is in the model. The HMC arm at $N = 10^4$ adds: the σ_u collapse in B1 under Pathfinder is a Pathfinder artifact, not a property of the data; the predicted GLM-scale slope on $\log(a_{\text{final}})$ from the link-translated α_{det} form lands inside the empirical 95% credible interval; and the conjecture's exhaustion content (B4 σ_u vanishes given a_{final} at $k = 6$) is confirmed at the predictive level.

The head-to-head does not show: Conjecture 3.1 is proven (it is not – the structural-empirical relationship is conjectured, supported by §4 and §5 but not derived); absolute log-score precision at the Bonacorsi-Bordoni production scale (the HMC arm is at $N = 10^4$, used as a structural-validation arm; absolute log-score numbers at $N = 10^7$ require HMC on production infrastructure that bypasses the Stan 2.36 lockup); a direct head-to-head comparison against the Bonacorsi-Bordoni reported number $-272,911.95$ under fully matched inference (their figure is HMC at full N in their NB2-GLM with mod-8 RE; B3 in this paper is Pathfinder at $N = 10^7$ with a_{final} FE at $k = 6$, and the 477-nat raw gap mixes inference method with model specification).

The conjecture's content is independent of inference method. The Pathfinder VI head-to-head and the HMC-at- $N = 10^4$ validation are two pieces of supporting evidence among several; the algebraic structure of §3, the per-class statistics of §4, and the 40-cell bridge to Tao (5.15) are the others.

6. Discussion

6.1. *What the conjecture would imply if true*

If Conjecture 3.1 is correct, the prefix decomposition's α_{det} is the algebraic content of empirically fit residue-class random effects in hierarchical Bayesian models on Collatz dynamics. The Bonacorsi-Bordoni mod8 random effect would not be a free hierarchical parameter to estimate, but a fixed effect derivable in closed form from the residue $r \bmod 8$. This has practical consequences for hierarchical Bayesian modeling on Collatz-like dynamics. At modular resolution k , a hierarchical model with a residue-class random effect estimates 2^{k-1} parameters (one per odd class), regularized by a hyperparameter τ . Under the conjecture, the structural content of those 2^{k-1} parameters is captured by at most $k + 1$ discrete α_{det} levels (one per

prefix-step count), a logarithmic reduction in parameter count. The model becomes more parsimonious without losing predictive content.

It also has interpretive consequences. A random effect in a hierarchical Bayesian model is an estimated pooling structure; its interpretation is “this level of hierarchy matters for prediction.” A fixed effect on α_{det} is closed-form: it identifies which residue-class structure is responsible for per-class variation, namely the Collatz prefix’s terminal multiplier a_{final} together with the prefix step count.

6.2. *Natural follow-up: HMC at full N*

The HMC validation arm at $N = 10^4$ (§5) confirms the conjecture’s structural content at the predictive level: σ_u in B1 is real (0.097, not a Pathfinder artifact), B3 outperforms B2 by more than B2 outperforms B1 (finer modular resolution captures additional structure), and σ_u in B4 vanishes (0.0075) once a_{final} at $k = 6$ is in the model (exhaustion confirmed). What remains as natural follow-up is HMC at the Bonacorsi-Bordoni production scale ($N = 10^7$ on their infrastructure), which would tighten absolute log-score precision and place the B3/B4 specifications on a directly comparable footing with their reported figure $-272,911.95$. The Stan 2.36 unified-mode lockup that prevented full- N HMC in this paper is data-scale-driven and bypassable by parallel-chains mode at smaller N (used here for the validation arm) or by Bonacorsi-Bordoni’s existing HMC infrastructure at full scale. Under such a run, Conjecture 3.1 predicts: B3 beats B1 by a margin reflecting a_{final} ’s structural advantage over the hierarchical mod-8 RE, and σ_u in B4 remains small.

The Bonacorsi-Bordoni group is best positioned to run this test, having the production HMC infrastructure and prior expertise with the model.

6.3. *Natural follow-up: structural rigor*

The conjecture is currently empirical-and-structural. A rigorous proof would require characterizing the algebraic content of the random effect’s posterior in a hierarchical NB2-GLM via the prefix decomposition’s algebraic structure. This is open. The closest available analytical tool is the prefix iteration itself (§3.1), which gives the closed-form $\alpha_{\text{det}}(r)$ on σ -additive scale; connecting that to the posterior mean of $u_{n \bmod 8}$ in the fitted Bayesian model – through the link-function rescaling characterized empirically in §5.4 – is the analytical work.

6.4. *Connection to the $qx + 1$ generalization*

A companion analysis examines the $qx + 1$ generalization ($q \in \{5, 7, 9, 11\}$) with two threads: an empirical Cramér convergence-rate finding ($q^{-\theta} = 2^{1-\theta} - 1$, closed form, matched empirically at $q = 5$, Cramér regime $q > 4$) and a partial structural extension of the prefix decomposition’s bridge equations – will be reported separately; preliminary results in, entries dated 2026-05-01 onward [3]. The $3x + 1$ conjecture above is the contribution to the Bonacorsi-Bordoni framework specifically; whether

the analogous conjecture holds for $qx + 1$ hierarchical models is an open question for $q \neq 3$.

6.5. Limitations

The empirical work is at $N \leq 2^{32}$ and modular resolutions $k \leq 12$. The signal-to-noise ratio $\text{SD}(\text{resid})/\text{mean}(\text{SE})$ for the α_{det} predictions is in the band $0.91 - 0.99$ across $k \in \{5, \dots, 12\}$; at $k = 4$ it is 0.70 (residuals smaller than per-class SE – prefix prediction over-explains at coarse resolution). At every k tested, the number of distinct a_{final} values equals k . Higher k might in principle reveal residual structure, but the trend across the nine resolutions tested is unambiguous: residuals scale with sampling noise at every k . The §5 head-to-head uses Pathfinder VI at $N = 10^7$ and HMC at $N = 10^4$; the HMC arm validates the structural ordering, the Pathfinder σ_u collapse, and the conjecture’s exhaustion content at the predictive level. HMC at the Bonacorsi-Bordoni production scale ($N = 10^7$) is the natural follow-up for tightening absolute log-score precision.

Conjecture 3.1 identifies $\alpha_{\text{det}}(r) = \text{prefix_steps}(r) + K_h \cdot \log(a_{\text{final}}(r)/2^k)$ as the algebraic content of the mod8 random effect at $k = 3$, on σ -additive scale. The corresponding random effect u_r in the NB2-GLM is $\alpha_{\text{det}}(r)$ rescaled by the link function, with leading-order coefficient $1/\mathbb{E}[\sigma]$ on the log-mean scale; the two-scale distinction is essential, and a literal comparison of u_r with $\log(a_{\text{final}}(r))$ on the same scale produces a magnitude discrepancy of order $\mathbb{E}[\sigma]$ that is purely a unit-conversion artifact. The “up to a global additive constant” caveat is essential: the conjecture does not predict the random effect’s absolute values, only its functional form as a function of $r \bmod 8$. The constant depends on the model’s overall normalization and is not load-bearing for the structural claim. The B2 fixed-effect dummies under HMC at $N = 10^4$ produce an empirical OLS slope on $\log(a_{\text{final}})$ of 0.0774 with 95% credible interval $[0.068, 0.087]$, against link-translated predictions of 0.0734 for the full α_{det} form (inside the interval) and 0.0675 for the $\log(a_{\text{final}})$ -only form (at the lower edge); the prefix-step term in α_{det} is exactly collinear with $\log(a_{\text{final}})$ across odd residues mod 8 and contributes the residual $\sim 6\%$ of the empirical slope above the $\log(a_{\text{final}})$ -only prediction. This is consistent with the conjecture’s α_{det} form.

7. Conclusion

This paper proposes a structural origin for the mod8 residue-class random effect introduced empirically by [1] for hierarchical Bayesian modeling of Collatz stopping times. The proposal is a conjecture: the mod8 random effect is, up to a global additive constant, the prefix decomposition’s per-class structural offset $\alpha_{\text{det}}(n \bmod 8) = \text{prefix_steps}(n \bmod 8) + K_h \cdot \log(a_{\text{final}}(n \bmod 8)/2^k)$ at $k = 3$, rescaled by the GLM’s log link with leading-order coefficient $1/\mathbb{E}[\sigma]$ on the log-mean scale.

The conjecture is supported by evidence on two sides. On the algebraic side, the prefix decomposition’s α_{det} predicts per-class statistics of σ across modular resolutions $k \in \{6, 7, 8, 9\}$ with $R^2 \geq 0.985$ and a signal-to-noise ratio

$\text{SD}(\text{resid})/\text{mean}(\text{SE}) \in [0.91, 0.99]$; the per-class variance, kurtosis, and skewness cluster by a_{final} at every tested k ; and the bridge to Tao 2022's (5.15) leading term holds at slope unity across 40 verification cells without parameter tuning. On the data side, a head-to-head Bayesian replication in the Bonacorsi-Bordoni framework runs in two arms: Pathfinder VI [8] at $N = 10^7$ and HMC at $N = 10^4$. The HMC arm at $N = 10^4$ confirms that B1's σ_u posterior is real (0.097, against the Pathfinder-collapsed 0.003*), that B3 outperforms B2 by more than B2 outperforms B1 (finer modular resolution captures additional structure), and that σ_u in B4 vanishes (0.0075) once a_{final} at $k = 6$ is in the model – the predictive-level confirmation of Conjecture 3.1's exhaustion content. B2's posterior coefficients on a_{final} at $k = 3$ produce an empirical OLS slope on $\log(a_{\text{final}})$ that contains the full α_{det} link-translated prediction inside its 95% credible interval, consistent with the conjecture once the prefix-step term is included.

If Conjecture 3.1 is correct, the prefix decomposition's α_{det} supplies the algebraic identity of the empirical random effect, replacing the mod8 hierarchical structure with a fixed effect on α_{det} – a logarithmic parameter count in the modular resolution. The joint model is what's interesting: the Bonacorsi-Bordoni predictive engine plus the prefix decomposition's closed-form covariate.

The conjecture is not proven. The Pathfinder VI head-to-head and the HMC-at- $N = 10^4$ validation are pieces of supporting evidence among several, alongside the algebraic structure of §3, the per-class statistics of §4, and the 40-cell bridge to Tao (5.15). HMC at the Bonacorsi-Bordoni production scale ($N = 10^7$) is the natural follow-up for tightening absolute log-score precision; the structural-empirical content above is independent of inference method.

Acknowledgments

The author thanks the maintainers of the Stan and cmdstanpy projects for the inference infrastructure used in §5, and the authors of the GHCN-D and OEIS data resources referenced throughout. The author is solely responsible for all claims, derivations, and computational results presented in this paper.

Use of AI assistance. The author used Claude (Anthropic) for LaTeX preparation, code review, and as a sounding board during analysis. Claude was not used to generate mathematical conjectures, perform original derivations, or write the final text of any analytical claim. All research design, empirical experiments, and conclusions are the author's; the author audited all computational results and takes full responsibility for the content of the paper.

Funding. The author received no external funding for this work.

Disclosure statement. The author reports no competing interests.

Data availability statement. All code, raw experimental output, and analysis scripts supporting the findings of this paper are available at <https://github.com>.

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com/mrnathanhumphrey-droid/Collatz, with a versioned archive on Zenodo at <https://doi.org/10.5281/zenodo.20100517>. Stan model files, fit objects, per-class regression outputs for §4, head-to-head replication artifacts for §5, and the 40-cell Tao bridge verification cells are all included. The repository is structured so that each numerical claim in §4 and §5 maps to a specific output file; a path index is provided in the repository’s `README.md`.

Appendix A: Computational artifact index

For reproducibility and referee verification, every numerical claim in §4 and §5 maps to a specific output file in the public repository (<https://github.com/mrnathanhumphrey-droid/Collatz>). The table below provides the mapping.

Section	Claim	Repository path
§3.3	Closed form $\langle \alpha_{\text{det}} \rangle = \log(6)/\log(4/3)$	<code>closed_form_findings.md</code>
§3.4	Bridge to Tao 2022 (5.15), closed form	<code>tao_bridge_findings.md</code>
§4.1	$k = 6$ hierarchical Bayesian fit, $R^2 = 0.9996$	<code>writeup.md</code> (Result 3)
§4.1	Table 1 k-sweep extension	<code>writeup.md</code> (Result 3 extension)
§4.2	Per-class moments cluster by a_{final}	<code>writeup.md</code> (Result 3, “the prefix de- termines all moments”)
§4.3	40-cell Tao bridge verification	<code>experiments/35/; writeup.md</code> (Result 3)
§4.4	Slope universality, $\tau_\beta = 0$ floor	<code>writeup.md</code> (Result 1)
§4.5	Tail-shape GPD ξ trend to- ward zero	<code>writeup.md</code> (Result 2)
§4.6	$\mu_\beta(N)$ oscillation, per-octave structure	<code>writeup.md</code> (Result 1)
§5.1	Pathfinder VI fit at $N = 10^7$	<code>experiments/06b/</code>
§5.1	HMC validation at $N = 10^4$	<code>experiments/paper1_hmc_n10k_validation/</code>
§5.4	B2 coefficient vs. link-translated prediction	<code>paper1_b2_coefficient_check.md</code>
§5.4	Prefix-steps / $\log(a_{\text{final}})$ collinearity	<code>paper1_prefix_steps_confound_check.md</code>

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